Nicole Lee

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Education

University of California, Berkeley — Expected Graduation: May 2028
B.S. in Electrical Engineering and Computer Science, B.S. in Bioengineering — Berkeley, CA

- **Hong Kong Scholarship for Excellence Scheme Scholar:** US$155,000 scholarship to study at UC Berkeley
- **Relevant Coursework:** Data Structures, Algorithms, Databases, Human-Centered Interfaces, Artificial Intelligence, Machine Learning
- **Activities:** Notion Campus Leader, Full-Stack Development Head TA (100+ students), Cal Hacks Director, Society of Women Engineers

Skills

Languages: TypeScript, JavaScript, Python, SQL, Swift, Go, Rust, C/C++, Java
Frontend: React, Next.js, React Native, GraphQL, React Three Fiber / WebGL, SwiftUI, Tailwind
Backend & Infra: Node.js, Express, FastAPI, PostgreSQL, Redis, MongoDB, Kafka, Docker, AWS (EC2, S3, Lambda), CI/CD
AI/ML: PyTorch, Hugging Face, LangGraph, MCP, RAG, vector search (pgvector, FAISS), LoRA fine-tuning, YOLOv12

Experience

Apple | Software Engineer Intern — May 2026 – Aug 2026

- Shipped a full-stack **MCP benchmarking system** into Apple's agent developer platform (React, TypeScript, PostgreSQL) that generates 200+ schema-aware queries per run across 450+ servers, cutting each run **18x** (6+ hrs → 20 min)
- Built the run-comparison UI, trace/diff viewers, failure-clustering dashboards, and config builder, with **virtualized tables** for large traces
- Architected a fault-tolerant **tracing proxy** (PostgreSQL state) with hybrid scoring (tool-call/JSON diffing + LLM-as-judge), collapsing 100+ correlated failures into <10 root causes, cutting debugging time **40%**

Cal Hacks | Director — Aug 2024 – Present

- Shipped the official **iOS app** (React Native, **10K+ installs**) with a 3D venue map (React Three Fiber, Gaussian splatting) embedded across 3 surfaces; GPU-profiled it to cut draw calls **74%** and triangles **84%** via instancing and render-on-demand
- Built the **Hive operations platform** (Next.js) for **4,000+ hackers** and 652 projects, with hardware checkout (**5,000+ orders**) and souvenir shop (3,500+ units) on transactional Postgres RPCs with row locking, 1-hr stock holds, and partial fulfillment
- Built an automated sponsorship outreach pipeline reaching 14,000+ contacts at 2,000+ companies, raising **$1,250,000+**

Wedge (YC S25) | Forward Deployed Engineer Intern — Aug 2025 – Dec 2025

- Built a clinician analytics web app for LA General Medical Center (600-bed Level I trauma center) in **React, TypeScript, and GraphQL**, surfacing EHR and scheduling data across 30K+ annual discharges to improve bed turnover
- Architected **20+ reusable React components** and REST data pipelines for FHIR/HL7 EHR data, adding client-side caching, conflict detection, and utilization scoring for low-latency analytics under concurrent workloads

Anchor Logics | Software Engineer Intern — Jun 2025 – Aug 2025

- Led 5 engineers to build **Propriologics**, a 0→1 telemetry platform (**Next.js, TypeScript, Node/Express**, AWS S3) that streams and visualizes 3D kinematic motion data from IoT sensor vests for **50+ ALS/Parkinson's patients**
- Fine-tuned a gait-analysis video pipeline (OpenPose, YOLOv12, DeepSORT, OpenCV), improving keypoint consistency **25%**, cutting occluded-frame tracking loss **35%**, and automating stride length, cadence, and step height

DigPath | Software Engineer Intern — Apr 2025 – Jun 2025

- Deployed a SageMaker + Meta SAM segmentation pipeline over 500+ whole-slide pathology images (70GB) on a serverless AWS Amplify/Lambda/DynamoDB platform, cutting annotation turnaround from days to under 2 hours per case

UCSF Memory and Aging Center | Undergraduate ML Researcher — Aug 2024 – Feb 2025

- Built a **RAG assistant** grounding GPT-4 Turbo answers in 2,000+ neurodegenerative-disease papers with full citations (author, year, journal, PMID), using overlapping chunking, MiniLM embeddings, and exact FAISS search

Projects

Aster [Live] [GitHub] | *TypeScript, Python, Next.js, FastAPI, SQLite, PyTorch, RDKit*

- Built an AI decision engine predicting which compounds drug-discovery labs should test next, combining a 5-model D-MPNN ensemble with Boltz-2 scoring; greedy selection found **27.3 more hits than random** over 10 seeds, benchmarked against 7 policies
- Built a reproducible **FastAPI/SQLite + Next.js** backend (content-hashed snapshots, two-phase batch commits, per-pick provenance) and ran a pre-registered study over **864 paired replays** across 9 targets with scrambled-evidence controls

PokerLab [GitHub] | *Swift, SwiftUI, Metal, SIMD, Swift Concurrency*

- Built a native iOS **game-theory-optimal** poker tutor on a custom Swift solver engine (hand evaluator, Monte Carlo equity engine, CFR/DCFR solvers, Metal GPU kernels) that computes **Nash equilibria on device in real time**
- Engineered a perfect-hash hand evaluator at **216M hands/s** (**4,000x** faster than brute force, verified on all 133M hands) and cut Nash solve time **29.6x** (2.3 s → 79 ms) with SIMD, multicore, and **custom Metal GPU kernels**

CodeMRI [GitHub] | *TypeScript, Python, FastAPI, Tree-sitter*

- Built a context engine for **coding agents** that parses repo-wide ASTs into persistent architecture and symbol graphs (call/import dependencies, change-impact analysis), compiling change-scoped context that cuts repo-reading tokens **83%**

Baymax's Cafe [Live] [GitHub] | *TypeScript, React, React Three Fiber, Vitest*

- Built a socially-aware robot navigation simulator that replans A* routes and 35–55 candidate paths in real time (**<10 ms p95**) around predicted human movement and learns user preferences through **active preference learning**